IGP TOPOLOGY

Goals:

1.) Create a 3 separate buildings with different dynamic routing protocol particular to OSPFv2, EIGRP, and RIPv2.

2.) Explore the hierarchy design of network devices such as layer 3 switch that act as distribution switch (distributing and routing the networks), where layer 2 switch act as access switch for end hosts.

3.) Apply what I have learned during the Jeremy's IT Labs lectures/laboratory activities such as PVSTP+,RPVST+, Etherchannel, VLSM, Subnetting, 802.1q, SVI, Inter-vlan, BPDU guard, Portfast, BGP (sample), and etc.

4.) Document it (PDF file format) if I accomplish the following goals.

Important commands:

Commands	Example	Purpose
do show cdp neighbor	R1(config)# do show cdp neighbor	To check router interface on which neighbor router interface they are connected to.
do show ip eigrp neighbor	R1(config)# do show ip eigrp neighbor	To verify if your router or other router are config as eigrp router
do show standby brief	R1(config)# do show standby brief	To check the configuration which network is ACTIVE/STANDBY in a router. It also includes virtual mac address and virtual gateway address
do show run | include ip dhcp excluded	R1(config)# do sh run | inc ip dh ex	To check the excluded range DCHP addresses and their DHCP pool name.
do show run | include ip route	R1(config)# do sh run | inc ip ro	To check the static route you have config in a router, since a router only show 1 default route at a time if conditions were met.

Interface port-number/0.vlan-id encapsulation dot1q 10	R1(config)# int g0/0.10 R1(config-subif)# encapsulation dot1q 10	To allow a router to put VLAN tag and enable one physical interface to handle multiple logical interface to do such ROAS topology
switchport mode access	SW1(config-if)#sw m ac	To set interface mode into ACCESS unconditionally
do show ip protocol	R1(config)# do sh ip pro	To check what kind of dynamic routing protocol is being used in internal autonomous system (AS)
ip dhcp excluded-address low ip address high ip address	R1(config)# ip dhcp excluded-address 192.168.0.1 192.168.0.10	To prevent a range of IPv4 address being given by DHCP to other hosts
no switchport	R1(config)# no sw	Used in layer 3 switch interface to act as layer 3 port to have an IP address and such
standby preempt	R1(config-if)# standby preempt	To retrieve the role that a router have lost due to a condition.
standby ip (the virtual gateway)	R1(config-if)# standby ip	To set the default gateway virtual IPv4 address
standby group vlan-id (not required)	R1(config-if)# standby group 10	To set which the group of an interface (using vlan-id which is optional)
standby version 2 (for IPv6)	R1(config-if)# standby version 2	To set version to use HSRP in IPv6 mode
standby version 1 (for IPv4)	R1(config-if)# standby version 1	To set version to use HSRP in IPv4 mode
router ospf process-id	R1(config)# router ospf 1	To configure which process id in OSPFv2 for a router
passive-interface interface-port	R1(config-router)# passive-interface g0/1	To supress/don’t receive OSPF hello messages or unnecessary traffic to end host intermediary device.
network address-network wild card bits	R1(config-router)# network 192.168.0.0 0.0.0.255	To enable OSPF in a particular interface
THERE’S MORE BUT ITS GONNA TAKE TOO LONG FOR ME TO COMPLETE IT. I should have document this while doing the project (lesson learned).		

What I have learned:

→ How to 802.1Q in a router to add VLAN tag which is similar to SVI in layer 3 switch

→ How to do OSPF in a building which consist of 4 different departments who have respective VLANs and applied Variable Length Subnet Mask (VLSM) for network distribution. I only set the building into OSPF Area 0 just for learning purposes.

→ How to enable and implement EIGRP in building B which I learn a lot from it since EIGRP for me is difficult to configure especially the complex computation of it but nonetheless I managed to ping the loopback address of an ISPR1 and eBGP it with building a network (192.168.0.0/25) all win for me.

→ How to enable and implement RIPv2 which for me is very easy to do and quite similar to OSPF with its configurations. I encounter a bug between the HSRP and EtherChannel but it was fixed in the next day which is hilarious for me since I kept thinking what's wrong between two collapsed routers.

→ How to do simple eBGP similar to a default route but in a manner for other ISPs to know each other known networks so end hosts can ping each other through eBGP. I'm not sure how theoretically it works, I just barely scratch the surface (configuration).

→ How to set HSRP for gateway redundancy which also help for DHCP server redundancy.

→ MOST IMPORTANTLY, I learn to become more persistent to learn up to this point and finish this big project and document it (not full details), just based on what I have done and understand.

Verification:

→ All PCs within the same building can ping each other (internal network)
→ All PCs default gateway are correct based on DHCP server
→ The dynamic routing protocol (OSPF, RIPv2, EIGRP) is implemented well which work fine for internal network to exchange data.
→ The point-to-point (P2p) connection between routers are working too (but some are not especially B-ISPR1)

Problems:

→ I need to 802.1Q first the routers in building A to create a SVI but works in single physical interface.

→ I need to be familiar how to set the OSPF especially opening an interface and what kind of area it will be used

→ I must ping the p2p connection between routers everytime now since most of my mistake is that misconfig mostly inter-router link especially in EIGRP, building B.

→ I have to understand the fundamentals once more (rinse and repeat) since most of what I have learned in the past is almost forgotten and not implemented into this project.

→ The 50% failed ICMP reply is the cause of improper configuration in ISP routers, especially the loopback address where it has to be static route to edge router to let other ISP know each other how to get their loopbacks.

→ I made a mistake in counting for subnetting which give me headache for DHCP server where I have to delete and add new DHCP configurations, likewise to DHCP excluded range IP address.

→ I don't understand how the access VLAN work and why it should one VLAN (im not sure myself so I just write this for future reference if I encounter this once again). THIS IS THE MAIN PROBLEM I DIDN'T UNDERSTAND.

→ The EIGRP is insanely hard for to implement for building B since there's no 'default-information originate' for EIGRP to share the static routes for collapsed routers

→ I struggle to RIPv2 because of the bug of allowed VLAN which UNTIL NOW I'm still stuck how to solve it (THIS IS MY NEXT PROJECT SO I CAN DOCUMENT THIS VERY WELL)

→ **etc...**